

Shayok Chakraborty

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Florida State University,
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PROFESSIONAL PREPARATION

Jadavpur University	Computer Science	B.S.	May 2006
Arizona State University	Computer Science	Ph.D.	Apr 2013

APPOINTMENTS

- **Associate Professor**, Department of Computer Science, Florida State University, 08/2023 – Present
- **Assistant Professor**, Department of Computer Science, Florida State University, 08/2017 – 08/2023
- **Assistant Research Professor**, School of Computing, Informatics and Decision Systems Engineering, Arizona State University, 2015 – 2017
- **Associate Director**, Center for Cognitive Ubiquitous Computing (CUbiC), Arizona State University, 2015 – 2017
- **Postdoctoral Research Associate**, Electrical and Computer Engineering, Carnegie Mellon University, 2014 – 2015
- **Postdoctoral Research Scientist**, Intel Labs, Oregon, 2013 – 2014
- **Graduate Research Intern**, Machine Learning department, Microsoft Research, Redmond, Jan – Apr 2012
- **Graduate Research Assistant**, Department of Computer Science and Engineering, Arizona State University, 2007 – 2013
- **Graduate Teaching Assistant**, Department of Computer Science and Engineering, Arizona State University, 2007 – 2012
- **Application Programmer**, IBM India, 2006 - 2007

RESEARCH INTERESTS

Machine Learning, Computer Vision, AI for Healthcare

AWARDS AND HONORS

- **Developing Scholar Award**, Florida State University, 2024
- **Graduate Teaching Award**, Department of Computer Science, FSU, 2024
- **NSF CAREER Award**, 2022
- **Faculty Research Award**, Department of Computer Science, FSU, 2022
- **Top Poster Award** at the Kentucky Conference on Health Communication 2022 (resulted from an interdisciplinary collaborative project)
- **Best Interdisciplinary Research Award** at the Aging and Health Informatics Conference 2021 (resulted from an interdisciplinary collaborative project)
- Selected as part of the **Top 25% of Program Committee members at AAAI 2021**
- Promoted to an **IEEE Senior Member**, 2021
- **AWS Machine Learning Research Award (MLRA)** 2019
- **Amazon Research Award (ARA)** 2019
- Research work featured in the **Tallahassee Democrat**, Jun 2018
- **2017 IEEE Multimedia Best Department Article Award** for our paper “Person-Centered Multimedia Computing: A New Paradigm Inspired by Assistive and Rehabilitative Applications”
- Research work featured in **ASU News**, 2013, 2017
- Nominated for the **Best PhD Dissertation Award**, Department of Computer Science, Arizona State University, 2013
- Selected as a finalist for the **Lane Fellowship in Computational Biology** at Carnegie Mellon University, 2013
- Recipient of the **University Graduate Fellowship**, Arizona State University, 2012
- Featured as an **Outstanding Graduate Student** by the Graduate College at Arizona State University, 2012
- Awarded **Travel Grants** to attend NIPS 2009 and 2010, CVPR 2010 and the Machine Learning Summer School at the University of California, Santa Cruz in 2012

TUTORIALS AND TEACHING

TUTORIALS

“*Active Domain Adaptation for Visual Computing Applications*”, IEEE Winter Conference on Applications of Computer Vision (**WACV**) 2023 (Co-presenter: Dr. Hemanth Venkateswara)

“*Active Learning for Multimedia Computing: Survey, Recent Trends and Applications*”, ACM Multimedia Conference (**ACM MM**) 2020 (**Solo Tutorial**)

“Active Learning for Computer Vision”, IEEE Winter Conference on Applications of Computer Vision (**WACV**) 2019 (**Solo Tutorial**)

“Active Learning for Multimedia Content Analysis”, IEEE International Conference on Multimedia and Expo (**ICME**) 2013 (Co-organizers: Dr. Vineeth Balasubramanian and Dr. Sethuraman Panchanathan)

INSTRUCTOR OF RECORD

Graduate

Weakly Supervised Machine Learning (Spring 2024)

Department of Computer Science
Florida State University

Data Mining (Fall 2017, Fall 2018, Fall 2019)

Department of Computer Science
Florida State University

Advanced Data Mining (Spring 2023)

Department of Computer Science
Florida State University

Theory of Automata and Formal Languages (Spring 2021, Spring 2022)

Department of Computer Science
Florida State University

Data Mining (Fall 2016, Spring 2017)

Department of Computer Science and Engineering
Arizona State University

Undergraduate

Introduction to Data Mining (Fall 2021, Fall 2022, Fall 2023)

Department of Computer Science
Florida State University

Programming Languages (Spring 2018, Spring 2019, Spring 2020, Fall 2020, Spring 2021)

Department of Computer Science
Florida State University

Mathematical Foundations of Informatics (Spring 2011)

Department of Computer Science and Engineering
Arizona State University

SERVICE

DEPARTMENTAL SERVICE

- **Director of the MS Program in Computer Science** (Fall 2023 – Present)
 - Responsible for initial student advising
 - Responsible for IEP reporting
 - Responsible for graduate student applications reviewing and decision making
 - Responsible for clearing MS students for graduation
- Member of the Faculty Recruitment Committee (2022 - 2023)
- Member of the Graduate Student Admissions Committee (2018 - 2022)
- Member of the Faculty Evaluation Committee (2018 - 2019)
- Core Organizing Committee Member, CS Expo 2017 (2017)

PROFESSIONAL ACTIVITIES

Session Chair

- **Interpretability and Explainability** at the ACM Conference on Knowledge Discovery and Data Mining (**KDD**) 2021
- **Semi-Supervised Learning** at the ACM Conference on Knowledge Discovery and Data Mining (**KDD**) 2020
- **Vision Systems and Applications** at the IEEE Winter Conference on Applications of Computer Vision (**WACV**) 2019, 2021
- **Machine Learning for Human Behavior Understanding and Assisted Living** at the IEEE International Conference on Machine Learning and Applications (**ICMLA**) 2011

Tutorials Co-Chair

- IEEE Winter Conference on Applications of Computer Vision (**WACV**) 2024

Area Chair

- International Conference on Learning Representations (**ICLR**) 2024
- Neural Information Processing Systems (**NeurIPS**) 2023, 2024

- IEEE Conference on Computer Vision and Pattern Recognition (**CVPR**) 2022, 2023, 2024
- IEEE Winter Conference on Applications of Computer Vision (**WACV**) 2019, 2020, 2021, 2022, 2023

Senior Program Committee

- **AAAI** Conference on Artificial Intelligence 2022

Program Committee

- **AAAI** Conference on Artificial Intelligence 2019, 2020, 2021, 2023, 2024, 2025
- ACM Conference on Knowledge Discovery and Data Mining (**KDD**) 2018, 2019, 2020, 2021, 2022, 2023
- International Joint Conference on Artificial Intelligence (**IJCAI**) 2016, 2018, 2022, 2023
- ACM Multimedia Conference (**ACM-MM**) 2016, 2017, 2018
- IEEE International Symposium on Multimedia (**ISM**) 2016, 2017
- ACM Symposium on Applied Computing (**ACM SAC**) 2017, 2018
- International Conference on Distributed Multimedia Systems, Visual Languages and Sentient Systems (**DMSVLSS**) 2017

Journal Review

- IEEE Transactions on Pattern Analysis and Machine Intelligence (**TPAMI**)
- ACM Computing Surveys (**CSUR**)
- IEEE Transactions on Neural Networks and Learning Systems (**TNNLS**)
- IEEE Transactions on Knowledge and Data Engineering (**TKDE**)
- IEEE Transactions on Affective Computing (**TAFEC**)
- Pattern Recognition Journal (**PR**)
- Neuro-computing Journal
- IEEE Journal on Selected Topics in Signal Processing (**J-STSP**)
- IEEE Internet of Things Journal
- IEEE Transactions on Human-Machine Systems (**THMS**)
- International Journal of Pattern Recognition and Artificial Intelligence (**IJPRAI**)
- IEEE Transactions on Autonomous Mental Development (**TAMD**)
- Signal Processing: Image Communication
- Information Sciences Journal
- CSI Transactions on ICT

Conference Review

- British Machine Vision Conference (**BMVC**) 2024
- IEEE Conference on Computer Vision and Pattern Recognition (**CVPR**) 2019, 2020, 2021

- IEEE International Conference on Computer Vision (**ICCV**) 2019, 2021
- European Conference on Computer Vision (**ECCV**) 2018
- IEEE Winter Conference on Applications of Computer Vision (**WACV**) 2015, 2016, 2017, 2018
- IEEE International Conference on Multimedia and Expo (**ICME**) 2018
- ACM Conference on Knowledge Discovery from Data (**KDD**) 2014 (External Reviewer)
- IEEE International Conference on Image Processing (**ICIP**) 2016 (External Reviewer)

Proposal Review

- NSF Panelist 2022

STUDENT MENTORING

Current

- Debanjan Goswami, PhD (**Advisor**)
- Ronast Subedi, MS (**Advisor**)
- Varun Totakura, MS (**Advisor**)
- Md Shamim Seraj, PhD (**Co-Advisor**)
- Reza Alishahi, PhD (Committee Member)
- Sumanth Bommireddy, PhD (Committee Member)
- Katie Brodhead, PhD (Committee Member)
- Isis Chen, PhD (Committee Member)
- Tonmoy Dey, PhD (Committee Member)
- Trina Dutta, PhD (Committee Member)
- Uzoamaka Ezeakunne, PhD (Committee Member)
- Rituparna Khan, PhD (Committee Member)
- Amit K. Nath, PhD (Committee Member)
- Mehran Sadeghi, PhD (Committee Member)
- Shaeke Salman, PhD (Committee Member)
- Mukadder Sodek, PhD (Committee Member)
- Fatema Tabassum Liza, PhD (Committee Member)
- Liting Zhang, PhD (Committee Member)
- Hansong Zhou, PhD (Committee Member)
- Christopher Holder Montenegro, MS (Committee Member)

Graduated

- Subhankar Banerjee, PhD (**April 2024, Advisor**)
- Aditya Bhattacharya, PhD (**July 2023, Advisor**)
- Ankita Singh, PhD (**June 2023, Advisor**)
- Mohammad Alahmadi, PhD (Nov 2020, Committee Member)
- Ali Alshahrani, PhD (Jul 2021, Committee Member)
- Daniel Bis, PhD (Jul 2022, Committee Member)
- Meysam Ghaffari, PhD (Jul 2020, Committee Member)
- Muhib Khan, PhD (Jun 2022, Committee Member)
- Chris Mills, PhD (Apr 2019, Committee Member)
- Esteban Parra, PhD (Jun 2022, Committee Member)
- Hiranmayi Ranganathan, PhD (Apr 2018, Committee Member)
- Ahana Roy Choudhury, PhD (Apr 2021, Committee Member)
- Abdulkarim Malkadi, PhD (July 2023, Committee Member)
- Hemanth Venkateswara, PhD (March 2017, Committee Member)
- Zi Wang, PhD (Jun 2022, Committee Member)
- Vishwas Torvi, MS (**May 2018, Thesis Advisor**)
- Jinglun Cai, MS (Mar 2022, Committee Member)
- Spencer Cain, MS (July 2023, Committee Member)
- Meng-Han Chung, MS (Apr 2020, Committee Member)
- Javier Escobar-Avila, MS (Aug 2019, Committee Member)
- Ronald Franco, MS (July 2018, Committee Member)
- Xiaoxiao Gan, MS (Dec 2018, Committee Member)
- Brendan Gressel, MS (Dec 2022, Committee Member)
- Shaima Guediri, MS (Apr 2023, Committee Member)
- Shiva Chaitanya Gurralla, MS (July 2021, Committee Member)
- Tejas Hasarali, MS (July 2019, Committee Member)
- Grzegorz Kakareko, MS (Sep 2019, Committee Member)
- Mayur Jain, MS (Dec 2020, Committee Member)
- Sai Swanitha Kodeboyina, MS (Nov 2023, Committee Member)
- Humaira Mauni, MS (May 2023, Committee Member)
- Alanna Mottesheard, MS (Apr 2022, Committee Member)
- Matthew Norflus, MS (Jul 2021, Committee Member)
- Nilay Patel, MS (Nov 2019, Committee Member)
- Raghav Rathi, MS (Dec 2019, Committee Member)
- Ricardo Romero, MS (Apr 2019, Committee Member)
- Mohammad Javad Shooshtari, MS (Apr 2023, Committee Member)
- Kunal SinghaRoy, MS (Apr 2018, Committee Member)
- Abanoub Sman, MS (Apr 2023, Committee Member)
- Jordan Snow, MS (Apr 2018, Committee Member)

- Vishak Udupa, MS (Dec 2023, Committee Member)
- Luis Villamil, MS (Apr 2022, Committee Member)
- Randy Bruno-Piverger, BS (Apr 2019, Committee Member)
- Cameron Farzaneh, BS (Nov 2018, Committee Member)
- Colin MacRae, BS (Feb 2022, Committee Member)
- Jeret McCoy, BS (Nov 2022, Committee Member)

GRANTS & FUNDING

Title: Multimodal Deepfake Information Manipulation, Analysis and Detection

Role: Co-PI

Source: Council on Research and Creativity (CRC), FSU

Award Period: 06/2024 – 06/2025

Total Award Amount: \$99,663

Title: CAREER: Active Learning in the Real-World

Role: PI

Source: NSF

Award Period: 05/2022 – 04/2027

Total Award Amount: \$519,882

Title: Center for Research and Education on Aging and Technology Enhancement - CREATE V

Role: FSU Co-PI

Source: NIA

Award Period: 08/2022 – 07/2027

Total Award Amount: \$3,100,000

Note: This is a multi-university collaborative grant, with the total award amount of \$14.7M. FSU's share of the award is \$3.1M

Title: Deep Active Learning with Relative Label Feedback

Role: PI

Source: Amazon

Award Period: 05/2020 – 04/2021

Total Award Amount: \$73,097

Title: Active Learning with Imperfect Oracles

Role: PI

Source: Amazon

Award Period: 05/2020 – 04/2021

Total Award Amount: \$35,000 (Cash) \$25,000 (AWS Promotional Credits)

Title: The Adherence Promotion with Person-Centered Technology (APPT) Project: Promoting Adherence to Enhance the Early Detection and Treatment of Cognitive Decline

Role: Co-PI

Source: NIA

Award Period: 07/2019 – 06/2024

Total Award Amount: \$2,996,314

Title: Research Supplements to Promote Diversity in Health-Related Research for the Adherence Promotion with Person-centered Technology (APPT) Project: Promoting Adherence to Enhance the Early Detection and Treatment of Cognitive Decline

Role: Co-PI and Mentor

Source: NIA

Award Period: 06/2021 – 05/2023

Supplement Award Amount: \$200,821

Title: Adaptive Training of Representational Flexibility in STEM+C for Autistic Adolescents

Role: Co-PI

Source: NSF

Award Period: 08/2018 – 08/2020

Total Award Amount: \$299,310

Title: Active Learning for Computer Vision

Role: PI

Source: Intel

Award Period: 01/2018 – 06/2018

Total Award Amount: \$25,074

Title: Deep Active Learning for Class Imbalance Problems

Role: PI

Source: Council on Research and Creativity (CRC), FSU

Award Period: 05/2018 – 08/2018

Total Award Amount: \$20,000

PUBLICATIONS

PhD THESIS

“Batch Mode Active Learning for Multimedia Pattern Recognition”, School of Computing, Informatics and Decision Systems Engineering, Arizona State University, April 2013

JOURNALS / BOOK CHAPTERS / MAGAZINE ARTICLES

A. Singh, **S. Chakraborty**, Z. He, Y. Pang, S. Zhang, R. Subedi, M. Lustria, N. Charness, W. Boot, “*Predicting Adherence to Computer-based Cognitive Training Programs among Older Adults: Study of Domain Adaptation and Deep Learning*”, **JMIR Aging** 2024

J. Moon, F. Ke, Z. Sokolikj, **S. Chakraborty**, “*Applying Multimodal Data Fusion to Track Autistic Adolescents’ Representational Flexibility Development During Virtual Reality-Based Training*”, **Computers & Education: X Reality** 2024

Z. He, M. Dieciuc, D. Carr, **S. Chakraborty**, A. Singh, I. Fowe, S. Zhang, M. Lustria, A. Terracciano, N. Charness, W. Boot, “*New Opportunities for the Early Detection and Treatment of Cognitive Decline: Adherence Challenges and the Promise of Smart and Person-Centered Technologies*”, **BMC Digital Health** 2023

A. Singh, **S. Chakraborty**, Z. He, S. Tian, S. Zhang, M. Lustria, N. Charness, N. Roque, E. Harrell, W. Boot, “*Deep Learning-based Predictions of Older Adults’ Adherence to Cognitive Training to Support Training Efficacy*”, **Frontiers in Psychology** 2022

Z. He, S. Tian, A. Singh, **S. Chakraborty**, S. Zhang, M. Lustria, N. Charness, N. Roque, E. Harrell, W. Boot, “*A Machine-Learning Based Approach for Predicting Older Adults’ Adherence to Technology-Based Cognitive Training*”, **Information Processing and Management** 2022

D. Carr, S. Tian, Z. He, **S. Chakraborty**, M. Dieciuc, N. Gray, M. Agharazidermani, M. Lustria, A. Dilanchian, S. Zhang, N. Charness, A. Terracciano, W. Boot, “*Motivation to Engage in Aging Research: Are There Typologies and Predictors?*”, **The Gerontologist** 2022

M. Ghaffari, A. Srinivasan, X. Liu, **S. Chakraborty**, “*High-resolution Home Location Prediction from Twitter Activities using Consensus Deep Learning*”, **Social Network Analysis and Mining (SNAM) Journal** 2021

S. Chakraborty, T. Wu, E. Forzani, C. Whisner, D. Jackemeyer, “*Long-Term Resting Metabolic Rate Analysis -Towards Prediction Models for Resting Metabolic Rate Changes*”, **International Journal of Prognostics and Health Management (IJPHM)** 2021

R. Ramakrishnan, B. Nagabandi, J. Eusebio, **S. Chakraborty**, S. Panchanathan, H. Venkateswara, “*Deep Hashing Network for Unsupervised Domain Adaptation*”, **Domain Adaptation in Computer Vision with Deep Learning**, **Springer** 2020

H. Ranganathan, H. Venkateswara, **S. Chakraborty**, S. Panchanathan, “*Deep Active Learning for Image Regression*”, **Deep Learning Applications**, **Springer** 2019

A. Singh, **S. Chakraborty**, “*Deep Domain Adaptation for Regression*”, Development and Analysis of Deep Learning Architectures, **Springer** 2019

S. Panchanathan, M. Moore, H. Venkateswara, **S. Chakraborty**, T. McDaniel, “*Computer Vision for Augmentative and Alternative Communication*”, Computer Vision for Assistive Healthcare, **Elsevier** 2018

H. Venkateswara, **S. Chakraborty**, S. Panchanathan, “*Deep Learning Systems for Domain Adaptation in Computer Vision*”, IEEE Signal Processing Magazine (**SPM**), Special Issue on Deep Learning for Visual Understanding 2017

S. Panchanathan, **S. Chakraborty**, T. McDaniel et al., “*Enriching the Fan Experience in a Smart Stadium Using Internet of Things Technologies*”, International Journal of Semantic Computing (**IJSC**) 2017 (Special Issue on Best of ISM 2016)

S. Panchanathan, **S. Chakraborty**, T. McDaniel, R. Tadayon, “*Person-Centered Multimedia Computing: A New Paradigm Inspired by Assistive and Rehabilitative Applications*”, IEEE Multimedia (**MM**) 2016 (**2017 IEEE Multimedia Best Department Article Award**)

S. Panchanathan, **S. Chakraborty**, T. McDaniel, “*Social Interaction Assistant: A Person-Centered Approach to Social Interaction for Individuals with Visual Impairment*”, IEEE Journal on Selected Topics in Signal Processing (**J-STSP**) 2016

S. Chakraborty, Q. Sun, V. Balasubramanian, S. Panchanathan, J. Ye, “*Active Batch Selection via Convex Relaxations with Guaranteed Solution Bounds*”, IEEE Transactions on Pattern Analysis and Machine Intelligence (**TPAMI**) 2015

S. Chakraborty, V. Balasubramanian, S. Panchanathan, “*Adaptive Batch Mode Active Learning*”, IEEE Transactions on Neural Networks and Learning Systems (**TNNLS**) 2015

V. Balasubramanian, **S. Chakraborty**, S. Panchanathan, “*Conformal Predictions for Information Fusion: A Comparative Study of P-Value Combination Methods*”, Annals of Mathematics and Artificial Intelligence (**AMAI**) 2015

V. Balasubramanian, **S. Chakraborty**, S.S. Ho, H. Wechsler, S. Panchanathan, “*Active Learning using Conformal Predictions*”, **Morgan Kaufmann Publ.** (Elsevier) 2013

S. Chakraborty, V. Balasubramanian, S. Panchanathan, “*Generalized Batch Mode Active Learning for Face-based Biometric Recognition*”, **Pattern Recognition** 2013

CONFERENCES AND WORKSHOPS

C. Jiang, H. Zhou, X. Zhang, **S. Chakraborty**, “*FedAR: Addressing Client Unavailability in Federated Learning with Local Update Approximation and Rectification*”, European Conference on Machine Learning and Principles and Practice of Knowledge Discovery in Databases (**ECML PKDD**) 2024

L. Zhang, L. Xu, S. Motamed, **S. Chakraborty**, F. de la Torre, “*D3GU: Multi-target Active Domain Adaptation via Enhancing Domain Alignment*”, IEEE Winter Conference on Applications of Computer Vision (**WACV**) 2024

D. Goswami, **S. Chakraborty**, “*Active Batch Sampling for Multi-label Classification with Binary User Feedback*”, IEEE Winter Conference on Applications of Computer Vision (**WACV**) 2024

Md S. Seraj, A. Singh, **S. Chakraborty**, “*Semi-supervised Deep Domain Adaptation for Deepfake Detection*”, Workshop on Manipulation, Adversarial and Presentation Attacks In Biometrics (**MAP-A**) at IEEE Winter Conference on Applications of Computer Vision (**WACV**) 2024

D. Goswami, **S. Chakraborty**, “*Active Learning for Video Classification with Frame Level Queries*”, IEEE International Joint Conference on Neural Networks (**IJCNN**) 2023

S. Chakraborty, A. Singh, “*Active Sampling for Text Classification with Subinstance Level Queries*”, **AAAI** Conference on Artificial Intelligence 2022 (**accepted for oral presentation**)

S. Banerjee, **S. Chakraborty**, “*Deterministic Mini-batch Sequencing for Training Deep Neural Networks*”, **AAAI** Conference on Artificial Intelligence 2021

A. Yuan, **S. Chakraborty**, “*A Study of Deep Learning for Predicting Freeze of Gait in Patients with Parkinson’s Disease*”, IEEE International Conference on Machine Learning and Applications (**ICMLA**) 2020

A. Singh, **S. Chakraborty**, “*Deep Active Transfer Learning for Image Recognition*”, IEEE International Joint Conference on Neural Networks (**IJCNN**) 2020

S. Banerjee, **S. Chakraborty**, “*Budgeted Subset Selection for Fine-tuning Deep Learning Architectures in Resource-Constrained Applications*”, IEEE International Joint Conference on Neural Networks (**IJCNN**) 2020

S. Chakraborty, “*Asking the Right Questions to the Right Users: Active Learning with Imperfect Oracles*”, **AAAI** Conference on Artificial Intelligence 2020

M. Woodham, J. Hawkins, A. Singh, **S. Chakraborty**, “*When to Pull Starting Pitchers in Major League Baseball? A Data Mining Approach*”, IEEE International Conference on Machine Learning and Applications (**ICMLA**) 2019

A. Bhattacharya, J. Liu, **S. Chakraborty**, “*A Generic Active Learning Framework for Class Imbalance Applications*”, British Machine Vision Conference (**BMVC**) 2019

C. Mills, J. Escobar-Avila, A. Bhattacharya, G. Kondyukov, **S. Chakraborty**, S. Haiduc, “*Tracing with Less Data: Active Learning for Classification-Based Traceability Link Recovery*”, IEEE International Conference on Software Maintenance and Evolution (**ICSME**) 2019

S. Banerjee, **S. Chakraborty**, “*DeepSub: A Novel Subset Selection Framework for Training Deep Learning Architectures*”, IEEE International Conference on Image Processing (**ICIP**) 2019

A. Bhattacharya, **S. Chakraborty**, “*Active Learning with n-ary Queries for Image Recognition*”, IEEE Winter Conference on Applications of Computer Vision (**WACV**) 2019

V. Torvi, A. Bhattacharya, **S. Chakraborty**, “*Deep Domain Adaptation to Predict Freezing of Gait in Patients with Parkinson’s Disease*”, IEEE International Conference on Machine Learning and Applications (**ICMLA**) 2018

H. Ranganathan, H. Venkateswara, **S. Chakraborty**, S. Panchanathan, “*Multi-Label Deep Active Learning with Label Correlation*”, IEEE International Conference on Image Processing (**ICIP**) 2018

S. Chakraborty, “*Distributed Active Learning for Image Recognition*”, IEEE Winter Conference on Applications of Computer Vision (**WACV**) 2018

S. Chakraborty, J. Stokes, L. Xiao, D. Zhou, M. Marinescu, A. Thomas, “*Hierarchical Learning for Automated Malware Classification*”, IEEE Military Communications Conference (**MILCOM**) 2017

H. Ranganathan, H. Venkateswara, **S. Chakraborty**, S. Panchanathan, “*Deep Active Learning for Image Classification*”, IEEE International Conference on Image Processing (**ICIP**) 2017

H. Venkateswara, J. Eusebio, **S. Chakraborty**, S. Panchanathan, “*Deep Hashing Network for Unsupervised Domain Adaptation*”, IEEE Conference on Computer Vision and Pattern Recognition (**CVPR**) 2017

H. Venkateswara, **S. Chakraborty**, T. McDaniel, S. Panchanathan, “*Model Selection with Nonlinear Embedding for Unsupervised Domain Adaptation*”, Workshop on Knowledge-based Techniques for Problem Solving and Reasoning (**KnowPros**) at the Association for the Advancement of Artificial Intelligence (**AAAI**) 2017

S. Panchanathan, **S. Chakraborty**, T. McDaniel et al. “*Smart Stadium for Smart Living: Enriching the Fan Experience*”, IEEE International Symposium on Multimedia (**ISM**) (**Invited Paper**) 2016

H. Venkateswara, **S. Chakraborty**, S. Panchanathan, “*Nonlinear Embedding Transform for Unsupervised Domain Adaptation*”, Workshop on Transferring and Adapting Source Knowledge in Computer Vision (**TASK-CV**) at the European Conference on Computer Vision (**ECCV**) 2016

H. Ranganathan, **S. Chakraborty**, S. Panchanathan, “*Transfer of Multi-modal Emotion Features in Deep Belief Networks*”, **Asilomar** Conference on Signals, Systems and Computers 2016

H. Ranganathan, **S. Chakraborty**, S. Panchanathan, “*Multimodal Emotion Recognition using Deep Learning Architectures*”, IEEE Winter Conference on Applications of Computer Vision (**WACV**) 2016

S. Chakraborty, O. Tickoo, R. Iyer, “*Towards Distributed Video Summarization*”, ACM Multimedia Conference (**ACM-MM**) 2015

S. Chakraborty, V. Balasubramanian, A. Ravi Sankar, S. Panchanathan, J. Ye, “*BatchRank: A Novel Batch Mode Active Learning Framework for Hierarchical Classification*”, ACM Conference on Knowledge Discovery and Data Mining (**KDD**) 2015

S. Chakraborty, O. Tickoo, R. Iyer, “*Adaptive Keyframe Selection for Video Summarization*”, IEEE Winter Conference on Applications of Computer Vision (**WACV**) 2015

S. Chakraborty, J. Zhou, V. Balasubramanian, S. Panchanathan, I. Davidson, J. Ye, “*Active Matrix Completion*”, IEEE International Conference on Data Mining, (**ICDM**) 2013

S. Chakraborty, H. Venkateswara, V. Balasubramanian, S. Panchanathan, “*Active Batch Selection for Fuzzy Classification in Facial Expression Recognition*”, IEEE International Conference on Machine Learning and Applications (**ICMLA**) 2011

R. Chattopadhyay, **S. Chakraborty**, V. Balasubramanian, S. Panchanathan, “*Optimization-based Domain Adaptation towards Person-Adaptive Classification Models*”, IEEE International Conference on Machine Learning and Applications (**ICMLA**) 2011

S. Chakraborty, V. Balasubramanian, S. Panchanathan, “*Optimal Batch Selection for Active Learning in Multi-label Classification*”, ACM Multimedia Conference (**ACM-MM**) 2011

S. Chakraborty, V. Balasubramanian, S. Panchanathan, “*Dynamic Batch Mode Active Learning*”, IEEE Conference on Computer Vision and Pattern Recognition (**CVPR**) 2011

S. Chakraborty, V. Balasubramanian, S. Panchanathan, “*An Optimization Based Framework for Dynamic Batch Mode Active Learning*”, Workshop on Optimization for Machine Learning at Neural Information Processing Systems (**NIPS**) 2010

V. Balasubramanian, **S. Chakraborty**, S. Krishna, S. Panchanathan, “*Enhancing Social Interactions of Individuals with Visual Impairments: A Case Study for Assistive Machine Learning*”, Workshop on Machine Learning for Assistive Technologies at Neural Information Processing Systems (**NIPS**) 2010

V. Balasubramanian, **S. Chakraborty**, S. Panchanathan, “*Multiple Kernel Learning for Efficient Conformal Predictions*”, Workshop on New Directions in Multiple Kernel Learning at Neural Information Processing Systems (**NIPS**) 2010

S. Chakraborty, V. Balasubramanian, S. Panchanathan, “*Dynamic Batch Size Selection for Batch Mode Active Learning in Biometrics*”, IEEE International Conference on Machine Learning and Applications (**ICMLA**) 2010

V. Balasubramanian, J.Ye, **S. Chakraborty**, S. Panchanathan, “*Kernel Learning for Efficiency Maximization in the Conformal Predictions Framework*”, IEEE International Conference on Machine Learning and Applications (**ICMLA**) 2010

S. Chakraborty, V. Balasubramanian, S. Panchanathan, “*Learning from Summaries of Videos: Applying Batch Mode Active Learning to Face-based Biometrics*”, IEEE Conference on Computer Vision and Pattern Recognition (**CVPR**) Workshop on Biometrics, 2010

S. Chakraborty, V. Balasubramanian, S. Panchanathan, “*Batch Mode Active Learning for Biometric Recognition*”, SPIE International Conference on Biometric Technology for Human Identification, SPIE Defense, Security & Sensing 2010

S. Marcel, C. McCool, **S. Chakraborty**, V. Balasubramanian, S. Panchanathan et al. “*On the Results of the First Mobile Biometry (MOBIO) Face and Speaker Verification Evaluation*”, International Conference on Pattern Recognition (**ICPR**) 2010

V. Balasubramanian, **S. Chakraborty**, S. Krishna, S. Panchanathan, “*Human- Centered Machine Learning in a Social Interaction Assistant for Individuals with Visual Impairments*”, Symposium on Assistive Machine Learning for People with Disabilities at Neural Information Processing Systems (**NIPS**) 2009

V. Balasubramanian, **S. Chakraborty**, S. Panchanathan, “*Online Active Learning using Conformal Predictions*”, Workshop on Analysis and Design of Algorithms for Interactive Machine Learning at Neural Information Processing Systems (**NIPS**) 2009

V. Balasubramanian, **S. Chakraborty**, S. Panchanathan. “*Generalized Query by Transduction for Online Active Learning*”, IEEE International Conference on Computer Vision (ICCV), Workshop on Online Learning for Computer Vision 2009

V. Balasubramanian, **S. Chakraborty**, S. Panchanathan, “*Multiple Cue Integration in Transductive Confidence Machines for Head Pose Classification*”, IEEE Conference on Computer Vision and Pattern Recognition (CVPR) Workshop on Online Learning for Classification, 2008

G. Gupta, S.K. Saha, **S. Chakraborty**, C. Mazumdar, “*Document Frauds: Identification and Linking Fake Documents to Scanners and Printers*”, IEEE International Conference on Computing, Theory and Applications (ICCTA) 2007

SHORT PAPERS AND ABSTRACTS

Z. Sokolikj, F. Ke, **S. Chakraborty**, J. Moon, “*A Comparison of Deep Learning and Supervised Machine Learning Methods in RF tracking for Neurodivergent Learners*”, ISLS Annual Meeting, International Society of the Learning Sciences (ISLS) 2024

Z. Sokolikj, F. Ke, **S. Chakraborty**, J. Moon, “*Using Deep Learning to Track Representational Flexibility Development of Children with Autism in a Virtual World*”, International Conference on Information and Education Technology (ICIET) 2023

J. Moon, F. Ke, Z. Sokolikj, **S. Chakraborty**, “*Multimodal Data Fusion to Track Representational Flexibility of Adolescents with Autism Spectrum Disorder during Virtual Reality-Based Training*”, ISLS Annual Meeting, International Society of the Learning Sciences (ISLS) 2022

J. Moon, F. Ke, Z. Sokolikj, **S. Chakraborty**, “*Using Sequence Mining to Explore the Representational Flexibility Development of Adolescents with Autism Spectrum Disorder in Virtual Reality-based Flexibility Training*”, ISLS Annual Meeting, International Society of the Learning Sciences (ISLS) 2021

N. Charness, W. Boot, **S. Chakraborty**, Z. He, “*Machine Learning Approaches to Understanding and Predicting Patterns of Adherence*”, Gerontological Society of America (GSA) 2021

M. Dieciuc, D. Carr, Z. He, **S. Chakraborty**, N. Charness, M. Lustria, A. Singh, W. Boot, “*An Introduction to the Adherence Promotion with Person-Centered Technology Project*”, Gerontological Society of America (GSA) 2020

M. Lustria, Z. He, W. Boot, **S. Chakraborty**, N. Charness, D. Carr, A. Terracciano, “*You Know You Are Getting Old When...: Improving Adherence to Mobile-Based Cognitive Assessments*

Using AI-Based Tailored Reminders”, Kentucky Conference on Health Communication, Department of Communication, University of Kentucky, 2020

TECHNICAL REPORTS

V. Balasubramanian, **S. Chakraborty**, S. Panchanathan, “*Confidence Estimation in Pattern Classification: An Analysis with Head Pose Estimation*”, Technical Report T09-12, School of Computing, Informatics and Decision Systems Engineering, Arizona State University, 2009

PRESENTATIONS AND INVITED TALKS

- Machine Learning Expo, **Florida State University**, Apr 2023
- Center for Imaging Science, **Rochester Institute of Technology**, Apr 2022
- Department of Computer Science, **Marquette University**, Apr 2022
- Translational AI Center, **Iowa State University**, Apr 2022
- Machine Learning Expo, **Florida State University**, Apr 2019
- College of Medicine, **Florida State University**, Jun 2018
- Machine Learning Expo, **Florida State University**, Apr 2018
- Department of Statistics, **Florida State University**, Sep 2017
- Department of Computer Science, **Florida State University**, June 2017
- Department of Computer Science, **Portland State University**, May 2017
- School of Informatics, Computing and Cyber Systems, **Northern Arizona University**, Feb 2017
- Visual Technology Showcase at the IEEE International Conference on Image Processing (**ICIP**), 2016
- **GoogleX**, Mountain View CA, 2016
- **Carnegie Mellon University**, Department of Computational Biology, 2013
- **Duke University**, Department of Computer Science, 2013
- **Microsoft Research Redmond**, Machine Learning Department, 2012
- Seventh **Annual Machine Learning Symposium** organized by the **New York Academy of Sciences**, 2012
- **Multimedia and Vision Meeting** organized jointly by **IBM Research** and **Stevens Institute of Technology**, 2012
- IEEE International Symposium on Multimedia (**ISM**), 2012
- **Doctoral Symposia** of Computer Vision and Pattern Recognition (**CVPR**) 2013, ACM Multimedia (**ACM MM**) 2011, the Association for the Advancement of Artificial Intelligence (**AAAI**) 2011 and the SIAM Data Mining Conference (**SDM**), 2011